

Nadeem Theba

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience designing and optimizing production ETL/ELT pipelines, lakehouses, and cloud data platforms. Proficient in Python, Advanced SQL, PySpark, dbt, Airflow, and AWS/GCP infrastructures. Specialized in real-time CDC streaming, vectorized query execution, and schema enforcement.

TECHNICAL SKILLS

- **Programming & Querying:** Python (Pandas, PySpark), SQL, Advanced SQL (CTEs, Window Functions)
- **Data Engineering & Governance:** dbt Core, Apache Airflow, ETL/ELT Pipelines, Dimensional Modeling (SCD Type 2), Data Validation
- **Streaming & Infrastructure:** Apache Kafka, PySpark Streaming, Debezium CDC, Redis, AWS (S3, EMR, Glue Catalog, CloudWatch, IAM), GCP, Databricks, Terraform, Docker, CI/CD
- **Databases & Lakehouse Storage:** Snowflake, PostgreSQL, MySQL, DuckDB, Apache Iceberg, Delta Lake, Parquet

PROFESSIONAL EXPERIENCE

Data Engineer (Contract) | Kaival Technologies | Remote | 10/2023 - Present

- Optimized high-volume enterprise batch pipelines (~500GB) using DuckDB out-of-core vectorized processing, reducing peak RAM from 32GB+ to <4GB and cutting cloud compute costs by 25%.
- Engineered data validation gates using dbt tests and Pandera schema contracts, routing malformed records to Dead Letter Queues (DLQ) and maintaining 99.9% data freshness SLA.
- Automated legacy data ingestion by engineering modular Python ETL pipelines, migrating 250k+ relational records weekly into cloud storage and eliminating 12 hours of manual operational overhead.
- Executed cross-cloud migration of enterprise datasets to GCP BigQuery and AWS S3, refactoring batch pipelines and optimizing complex SQL joins across 50M+ rows to reduce query execution latency by 65%.

SELECTED TECHNICAL PROJECTS

AWS Real-Time Fraud Feature Store & Cloud Decision Engine | AWS S3, Apache Iceberg, Kafka, PySpark, Redis, Debezium, dbt

- Built a CDC streaming pipeline capturing PostgreSQL 16 WAL updates via Debezium and Apache Kafka, delivering sub-10ms online feature retrieval in Redis for real-time fraud scoring.
- Implemented PySpark Structured Streaming micro-batches with idempotent sink logic, materializing dual-write feature aggregations into an Apache Iceberg lakehouse and Redis online store.
- Designed dbt Core dimensional models incorporating SCD Type 2 history tracking on Apache Iceberg, utilizing Pandera contracts to enforce schema constraints for regulatory compliance.

AWS Predictive Supply Chain IoT Telemetry Lakehouse | AWS Kinesis, Databricks, PySpark, AWS S3, Terraform, dbt

- Architected an AWS streaming pipeline processing 50,000 IoT events/sec using a 50-shard Kinesis Data Stream, ingesting real-time telemetry into AWS S3 Delta Lake.
- Configured Databricks Auto Loader with incremental file discovery and physical partitioning by plant and date, achieving a 90% reduction in S3 scan volume for Databricks SQL queries.
- Deployed dbt Databricks models executing 24-hour rolling window aggregations for machine risk scoring, implementing automated quarantine routing to S3 DLQ for invalid telemetry streams.

AWS Coverdrive Cricket Telemetry Cloud Lakehouse | AWS S3, Delta Lake, PySpark, DuckDB, dbt, Apache Airflow, XGBoost

- Engineered an end-to-end telemetry pipeline processing ball-by-ball match streams into AWS S3 Delta Lake following Medallion Architecture (Bronze, Silver, Gold layers).
- Formulated partition-isolated feature scaling and PCA transformations fitted strictly on training partitions (match_id), eliminating target data leakage across inference pipelines.
- Developed an automated Apache Airflow DAG orchestrating daily batch prediction pipelines for a dual-target XGBoost model (0.7719 AUC-ROC), pushing probabilities to Gold layer storage.

EDUCATION

MSc Data Science & Analytics | University of Hertfordshire, UK (2021 - 2023)

BTech Computer Science | Marwadi University, India (2016 - 2020)